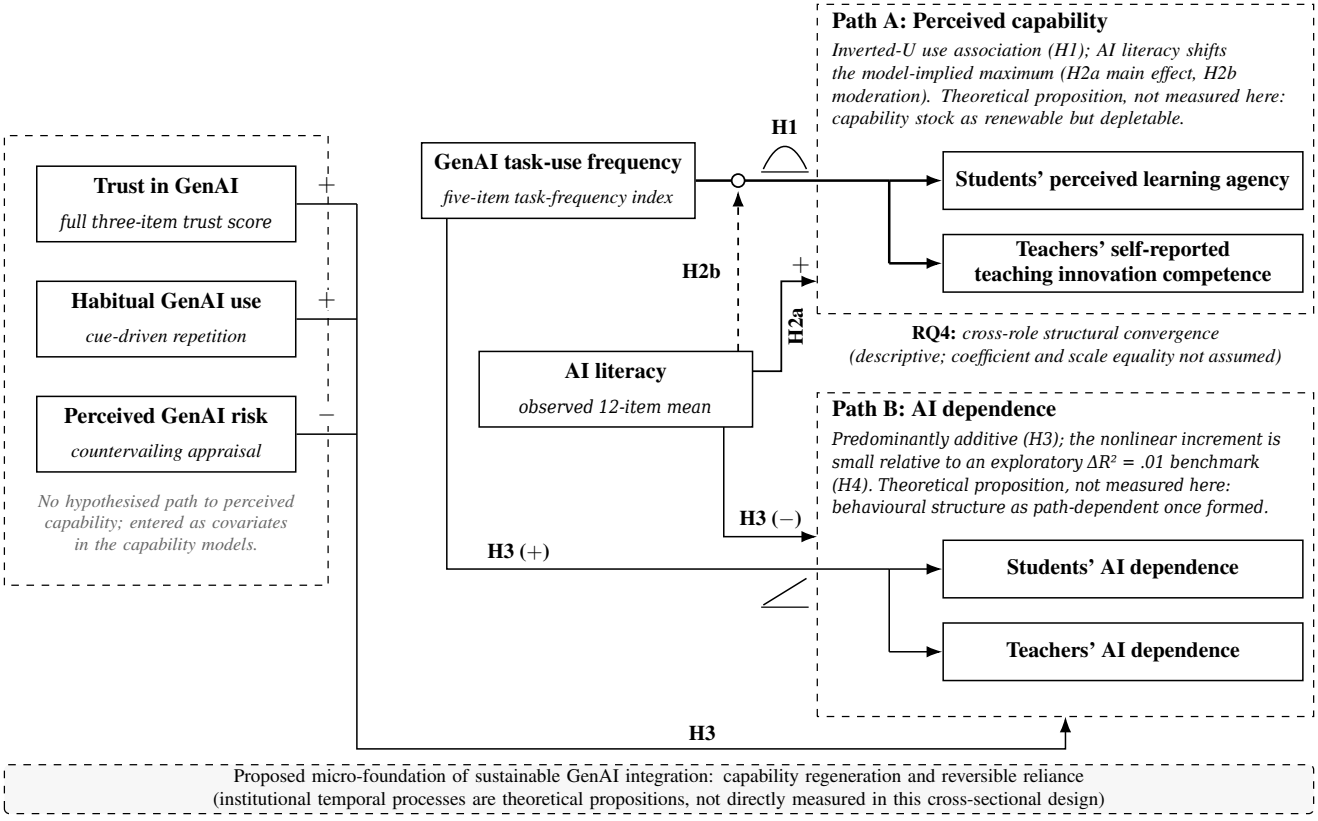

Path A: Perceived capability
Inverted-U use association (H1); AI literacy shifts the model-implied maximum (H2a main effect, H2b moderation). Theoretical proposition, not measured here: capability stock as renewable but depletable.
H1
GenAI task-use frequency
five-item task-frequency index
Students' perceived learning agency
Teachers' self-reported teaching innovation competence
H2b
H2a
RQ4: cross-role structural convergence
(descriptive; coefficient and scale equality not assumed)
Trust in GenAI
full three-item trust score
Habitual GenAI use
cue-driven repetition
Perceived GenAI risk
countervailing appraisal
No hypothesised path to perceived capability; entered as covariates in the capability models.
AI literacy
observed 12-item mean
Path B: AI dependence
Predominantly additive (H3); the nonlinear increment is small relative to an exploratory ΔR² = .01 benchmark (H4). Theoretical proposition, not measured here: behavioural structure as path-dependent once formed.
H3 (−)
H3 (+)
Students' AI dependence
Teachers' AI dependence
H3
Proposed micro-foundation of sustainable GenAI integration: capability regeneration and reversible reliance
(institutional temporal processes are theoretical propositions, not directly measured in this cross-sectional design)